Sri Lanka Institute of Information Technology

IT2011 – Artificial Intelligence and Machine Learning

Year 2 semester 1

Group 7

Lung Cancer Survival Prediction

Final Report

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Content

1. Introduction and Problem Statement

Lung cancer continues to be one of the leading causes of cancer-related deaths worldwide. Accurate and timely prediction of survival (recovery or death) may be employed to inform clinical treatment decisions and resource planning. This project compares and builds six supervised classification models to predict the binary outcome variable (survived).

These include

- Support Vector Machine (SVM)
- Logistic Regression
- Decision Tree
- Naive Bayes
- Random Forest
- k-Nearest Neighbors (k-NN).

The data is split into training (80 percent) and testing (20 percent) after preprocessing. Hyperparameter tuning is performed prior to final training.
Class imbalance is one of the central concerns: the positive class (surviving or recovered patients) is the minority and adversely affects various models. Class weighting is applied to combat imbalance; the optimum seen accuracy to date, however, is that of the Naive Bayes classifier.

2. Dataset Description

- Source : Lung Cancer Dataset

A structured tabular dataset with 17 columns and 100508 rows with one row per patient and the following columns:

- Id
- Age
- Diagnosis-date
- End-treatment-date
- Family-history
- BIM
- Cholesterol-level
- Hypertension
- Asthma
- Cirrhosis
- Other-cancer
- Gender-Female, gender-Male
- Country-Austria, Croatia, Czech Republic, Germany, Hungary, Italy, Malta, Netherlands, Poland, Sweden
- Cancer-stage-Stage I, Stage II, Stage III, Stage IV
- Smoking-status-Current Smoker, Former Smoker, Never Smoked, Passive Smoker
- Treatment-type-Chemotherapy, Combined, Radiation, Surgery
- Survived (binary target, indicates whether the patient recovered or died)

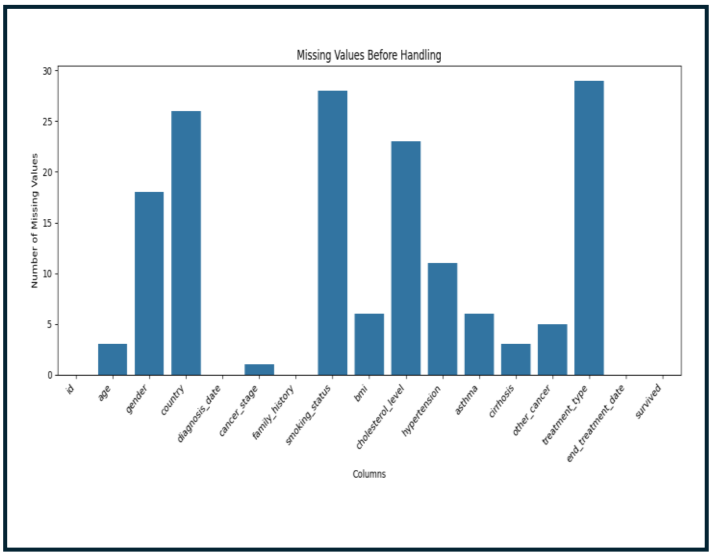
Variable Roles

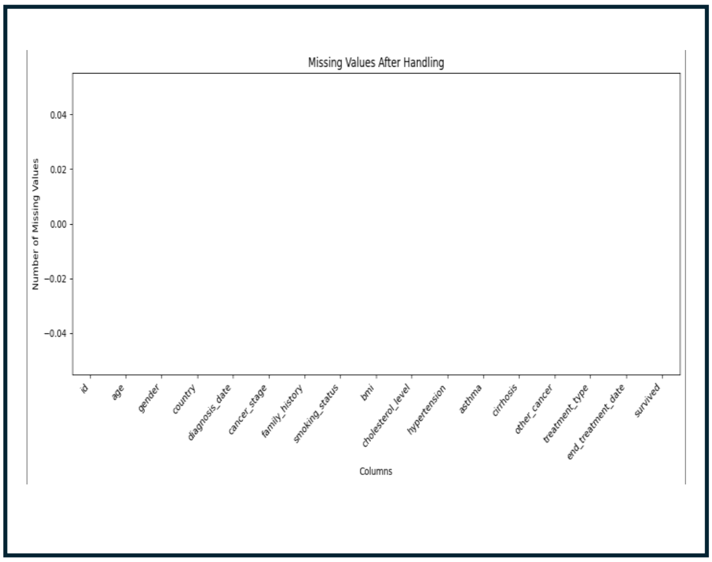
- Identifier: id (not used for learning)
- Dates: diagnosis-date, end-treatment-date (used to derive intervals or excluded as appropriate)
- Numerical features: age, bmi, cholesterol-level (and any engineered durations)
- Binary indicators: comorbidities, gender dummies, one-hot country, one-hot cancer stage, one-hot smoking status, one-hot treatment type
- Target: survived (It shows a high class imbalance where minority is the positive class)

3. Preprocessing and Exploratory Data Analysis (EDA)

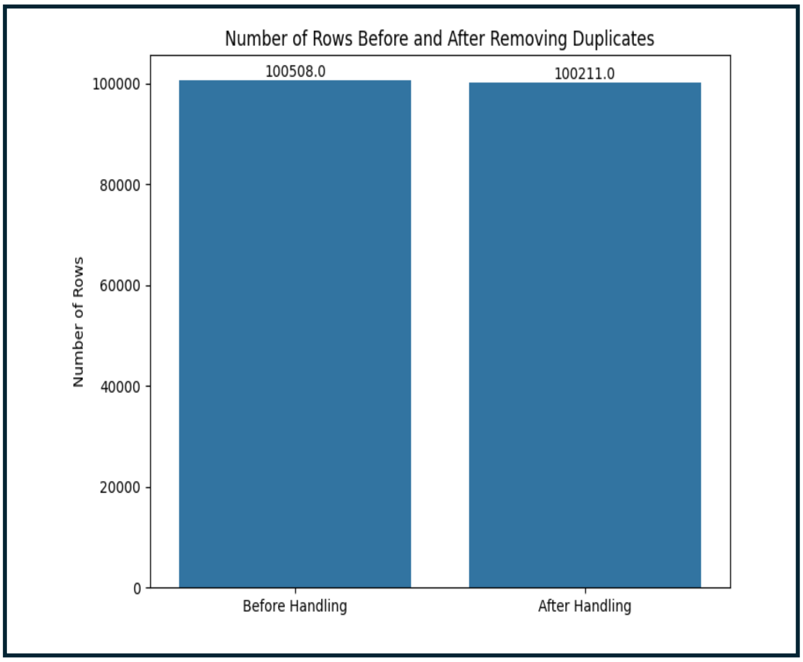
Data Quality and Cleaning

- Handled missing values using appropriate strategies (imputation consistent with data type and clinical plausibility).

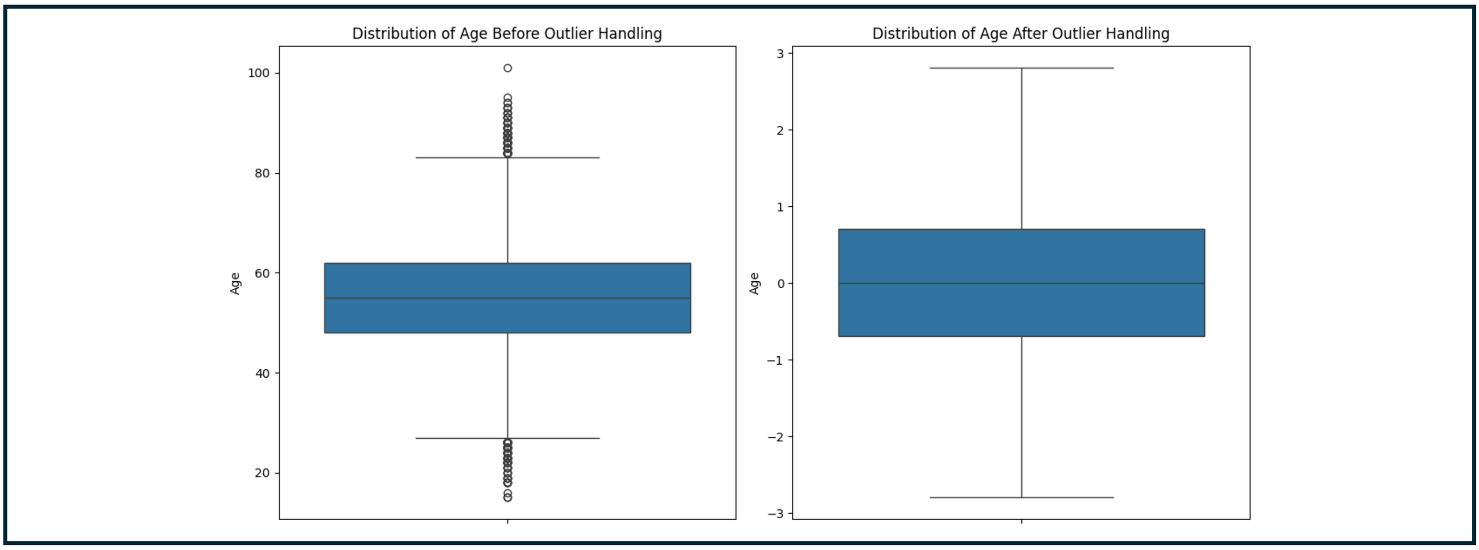




- Removed duplicate records after verifying identical identifiers or feature sets.

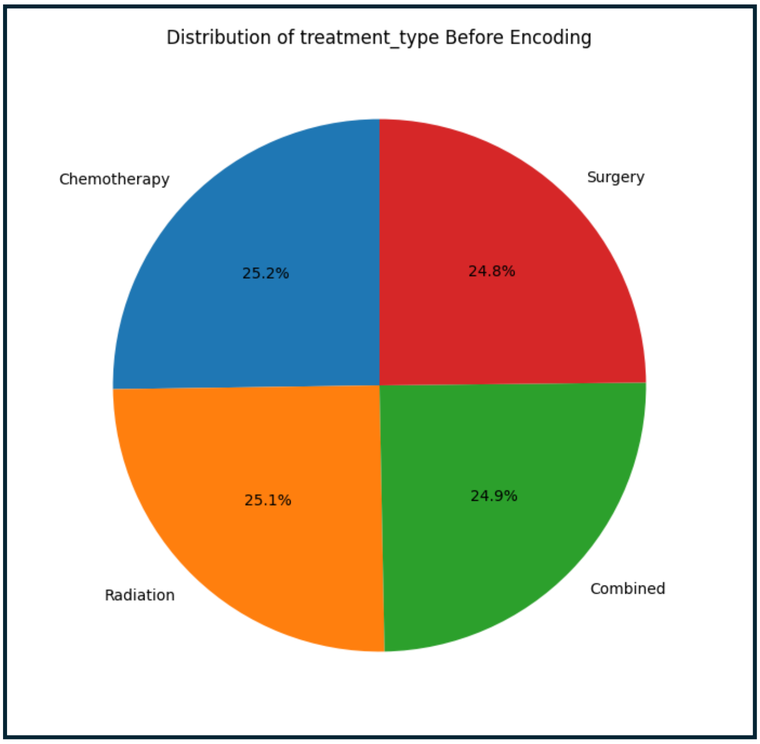


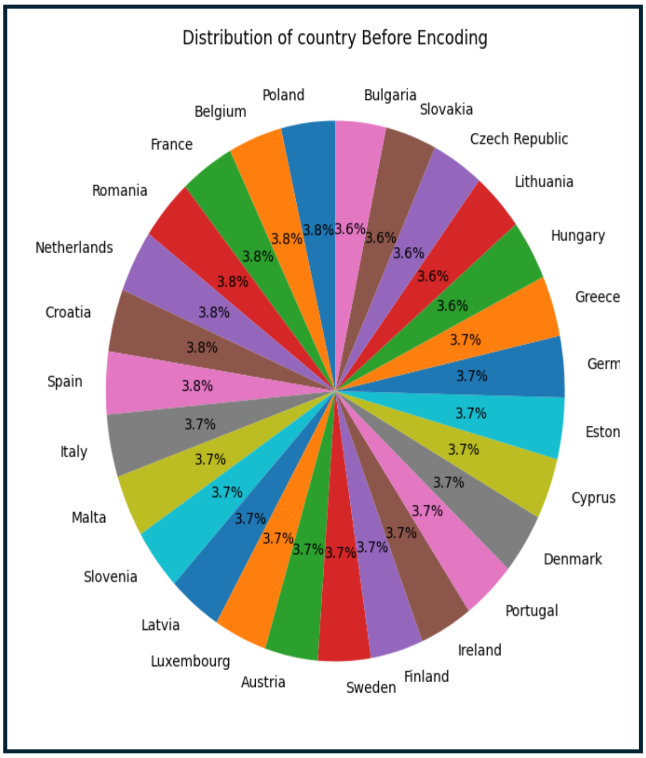
- Addressed outliers for numerical features (e.g., IQR-based capping) carefully to avoid suppressing clinically meaningful extremes.

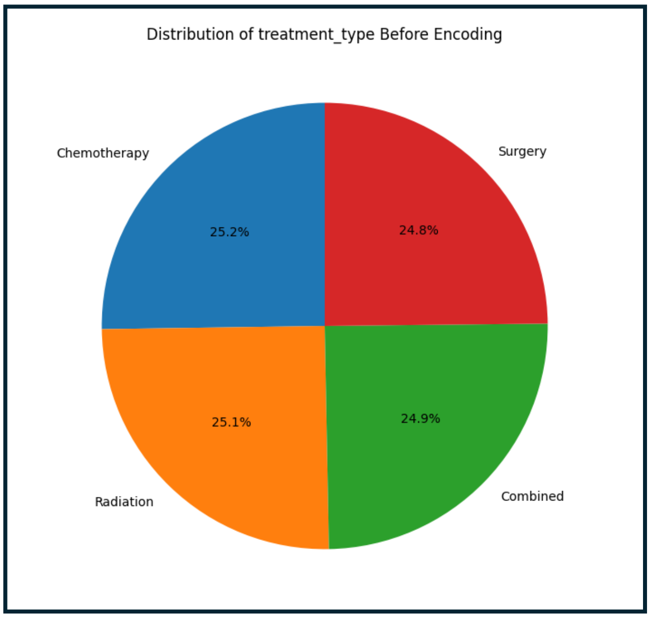


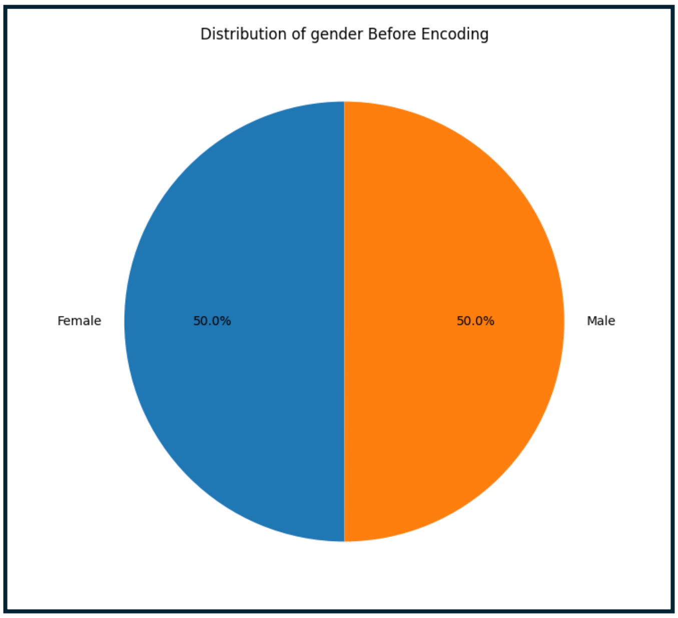
Feature Engineering and Encoding

- Encoded categorical variables using one-hot encoding (already present in dataset columns).

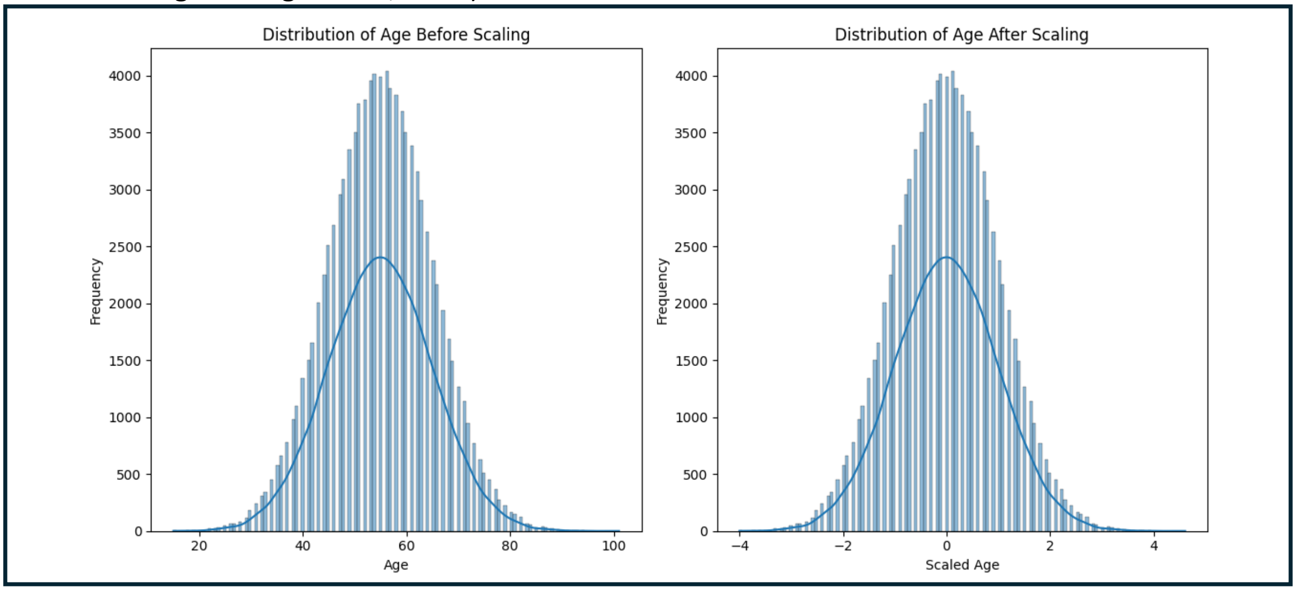




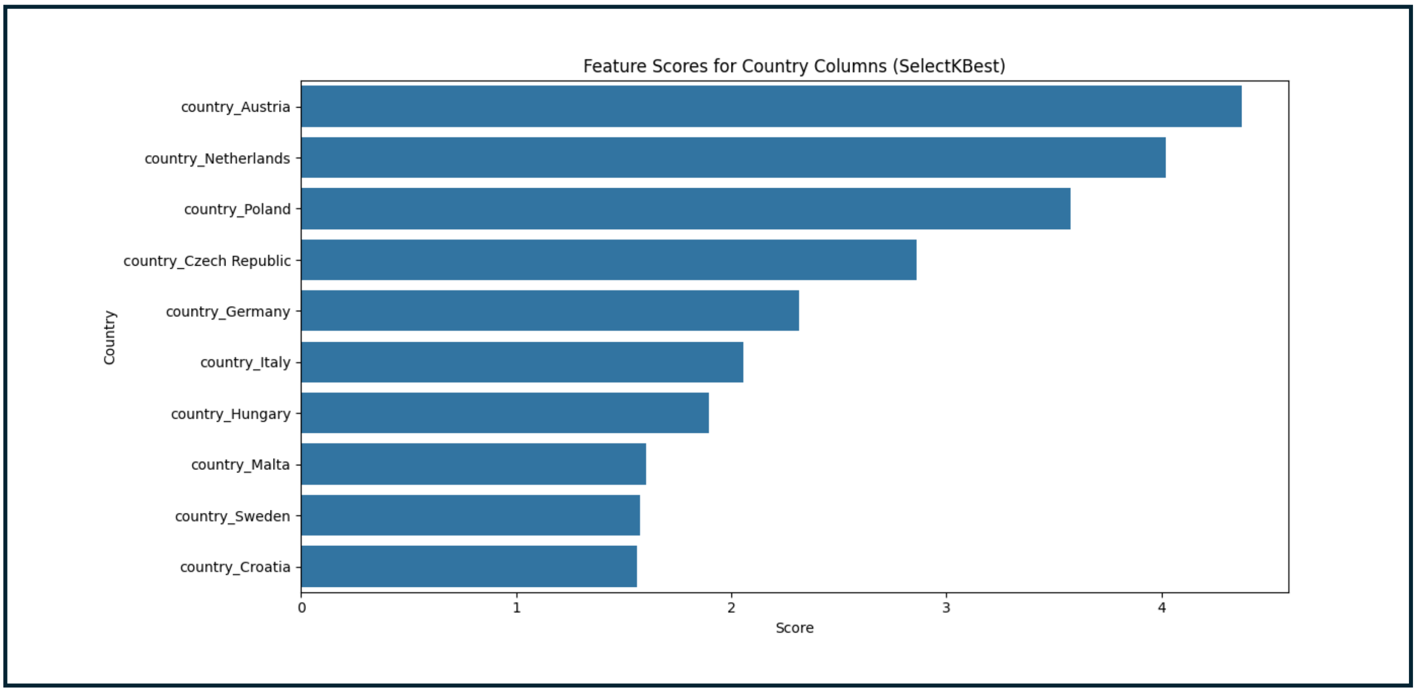




- Scaled numerical features (age, bmi, cholesterol-level, and any engineered durations) to standardize ranges for algorithms sensitive to feature scale (SVM, Logistic Regression, k-NN).



- . Feature selection: choose only the best scored country with respect to the dependent variable and remove unnecessary countries.



4. Model Design and Implementation

Data Split

- Train 80 percent and Test 20 percent, preserving class proportions via stratification to reflect the imbalance in both splits.

Training Pipeline

- For scale-sensitive models (SVM, Logistic Regression, k-NN), applied scaling within a pipeline to avoid data leakage during cross-validation and hyperparameter tuning.
- Hyperparameter tuning performed before final training (for example, Grid Search or Randomized Search with stratified folds), optimizing primarily for balanced performance metrics under class imbalance.
- Class weights were applied to minority positive class for algorithms that support it.

Models and Tuned Hyperparameters

1. Support Vector Machine (SVM) : IT24102922 (Ahangama A.V.D.S)

Model Architecture

- Separates data into two classes (survived = 0 or 1) using the **best line or curve**.
- Uses **kernels** (e.g., linear or RBF) to capture both simple and complex patterns.
- Focuses on key boundary points called **support vectors**.

Key Hyperparameters

- **C**: 0.1–10 (best: 1) → controls margin vs. misclassification.
- **kernel**: 'rbf' (best) → handles non-linear data.
- **gamma**: 0.001–1 (best: 0.1) → defines curve shape for RBF kernel.
- **class_weight**: 'balanced' → handles data imbalance (81% not survived).
- **scaler**: StandardScaler → improves accuracy.
- **random_state**: 42 → ensures consistent results.

Dataset Suitability

- Works well after **scaling and preprocessing** (remove dates, handle imbalance).
- Handles **mix of numeric and categorical data (~100k rows)** but may be **slow** on large datasets.

Overall Insight

- **Tuned SVM** improves cancer detection (recall ~35%) compared to basic models.

2. Logistic Regression : IT24103014 (Abeysinghe J.H.C.M)

Model Architecture

- Simple linear model predicting survival (0 or 1).
- Creates a straight decision boundary — no hidden layers or complex structure.

Key Hyperparameters

- **C**: 0.001–0.01 → controls regularization strength to prevent overfitting.
- **solver**: 'liblinear' → efficient for small to medium datasets.
- **max_iter**: 1000–2000 → limits training iterations.
- **class_weight**: 'balanced' or {0: 0.64, 1: 2.68} → gives more weight to minority class (survived = 1, 19%).
- **penalty**: 'l2' → adds regularization to model weights.
- **random_state**: 42 → ensures consistent results.

Dataset Suitability

- Around **100k samples** with numeric and categorical features.
- **Imbalanced** (81% not survived, 19% survived).
- Logistic Regression is suitable as a **baseline model** but not ideal for imbalanced data.

Overall insight

- Works well as a **starting model** for binary prediction.
- **Low recall** means many cancer cases are missed — risky for medical use.

3. Decision Tree Classifier : IT24103005 (Gunawardhana M.D.K)

Model Architecture

- Predicts survival (0 or 1) by splitting data into branches using feature-based questions (e.g., *age > 60?*).
- Forms a clear tree structure, easy to interpret but can overfit easily.

Key Hyperparameters

- max_depth: 5–20 (best: 10) → limits tree depth to control overfitting.
- min_samples_split: 2–10 (best: 5) → minimum samples to split a node.
- class_weight: 'balanced' → adjusts for data imbalance (81% not survived).
- criterion: 'gini' (best) → measures quality of splits.
- random_state: 42 → ensures consistent results.

Dataset Suitability

- Works well on mixed data (~100k rows) after preprocessing (e.g., removing date columns).
- Can easily overfit, especially with many features or imbalance.

<u>Overall Insight</u>

- Tuned Decision Tree improves recall to ~36%, performing better than Naive Bayes or default models.

4. Naive Bayes Classifier : IT24103084 (Karunanayake K.M.C.N)

<u>Model Architecture</u>

- Simple **probabilistic model** that predicts survival (0 or 1).
- Assuming all features (e.g., age, BMI) are **independent** and **normally distributed**.
- Uses probability formulas—no hidden layers or complex structure.

<u>Key Hyperparameters</u>

- **priors**: Based on class proportions (e.g., [0.81, 0.19]) to manage imbalance.
- **var_smoothing**: 1e-9 (default) → improves numerical stability.

<u>Dataset Suitability</u>

- Dataset: ~100k samples, numeric + categorical features, class imbalance (81% not survived, 19% survived).
- Naive Bayes works best for continuous, independent features, but struggles with mixed or correlated data.

<u>Overall Insight</u>

- Not suitable for lung cancer survival prediction.
- Performs poorly on imbalanced and categorical data.

5. Random Forest Classifier : IT24102935 (Rajamanthree R.M.K.V**)**

<u>Model Architecture</u>

- An ensemble model made of many decision trees that vote to predict survival (0 or 1).
- Each tree uses different parts of the data → combined results improve accuracy and reduce overfitting.
- More flexible and powerful than Logistic Regression or Naive Bayes.

<u>Key Hyperparameters</u>

- n_estimators: 50–200 (best: 50) → number of trees.
- max_depth: 10–20 (best: 10) → tree depth limit.
- min_samples_split: 2–10 (best: 5) → minimum samples to split a node.
- class_weight: 'balanced' → handles class imbalance (19% survived).
- random_state: 42 → ensures consistent results.
- scaler: StandardScaler → normalizes data.

Dataset Suitability

- Works well for large, mixed-feature datasets (~100k rows).
- Handles imbalanced data better than simpler models after preprocessing (e.g., removing date/family columns).

Overall Insight

- Tuned and shallow models improve recall up to 36%, detecting more positive cases.

6. k-Nearest Neighbors (k-NN) Classifier : IT24103049 (Ransith K.A.K)

Model Architecture

- Predicts survival (0 or 1) by checking the **k closest data points** in the dataset.
- Uses **distance measures** (like Euclidean) to find neighbors.
- No actual training — it just stores the data and compares during prediction.

Key Hyperparameters

- **n_neighbors**: 3–15 (best: 5) → number of neighbors.
- **weights**: 'distance' → closer neighbors have more influence.
- **metric**: 'minkowski' (p=2 = Euclidean distance).
- **scaler**: StandardScaler → ensures fair distance calculations.
- **random_state**: 42 → consistent data splits.

Dataset Suitability

- Works on **mixed data (~100k rows)** after preprocessing (remove date fields, scale features).
- Struggles with **large datasets** and **class imbalance (81% not survived, 19% survived)**.

Overall Insight

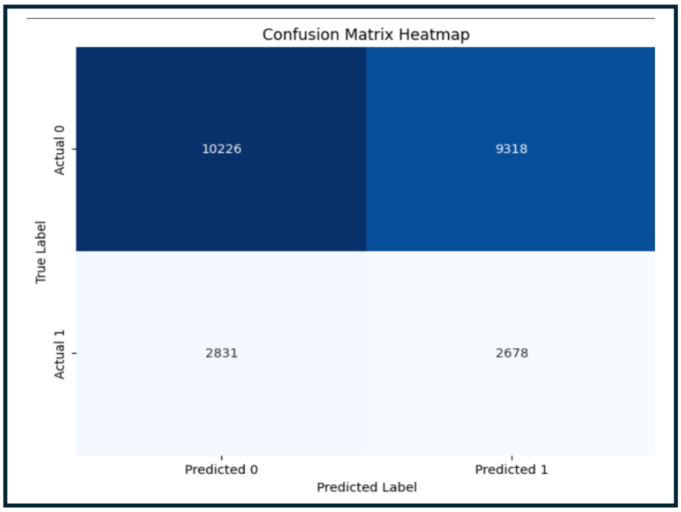
- Detects nearly half of positive cases (**recall ~49%**), better than Naive Bayes.
- However, precision is low (many false positives), and it's **slow for large datasets**.

5. Evaluation and Comparison

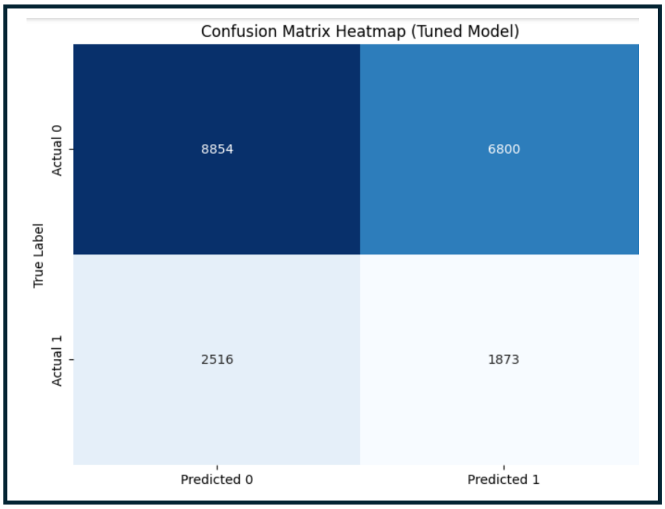
The abnormal decrease in model performance metrics is caused by the high imbalance of the dependent variable. Because one class heavily outweighs the other, the model tends to favor the majority class, making it difficult to achieve high accuracy or balanced predictions even with extensive training or tuning.

<u>Metrics</u>

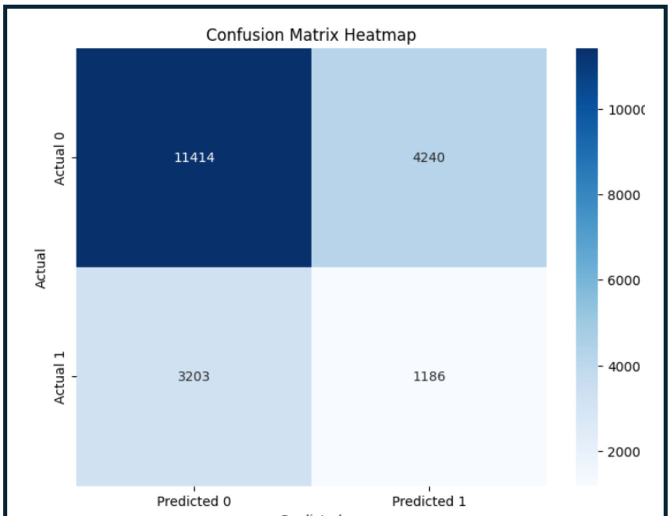
1. Logistic Regression
    - Without handling the class imbalance
        - Accuracy: 0.7801
        - Precision: 0.0000
        - Recall: 0.0000
        - F1-score: 0.0000
    - With handling class imbalances and hyper-parameter tuning
        - Accuracy: 0.5151
        - Precision: 0.2232
        - Recall: 0.4861
        - F1 Score: 0.3060



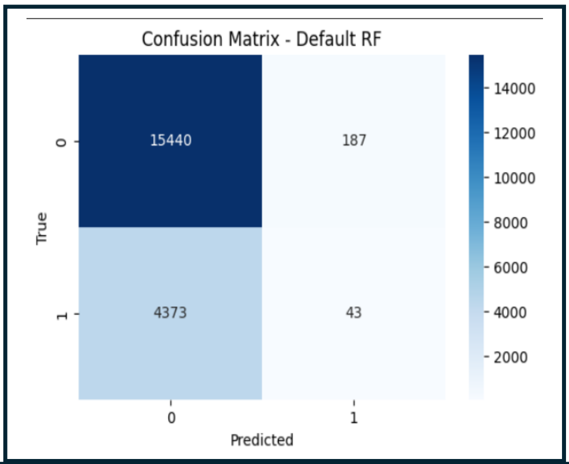
2. Decision Tree Classifier
    - After handling class imbalances and hyperparameter tuning
        - Accuracy: 0.5352
        - Precision: 0.2160
        - Recall: 0.4267
        - F1 Score: 0.2868



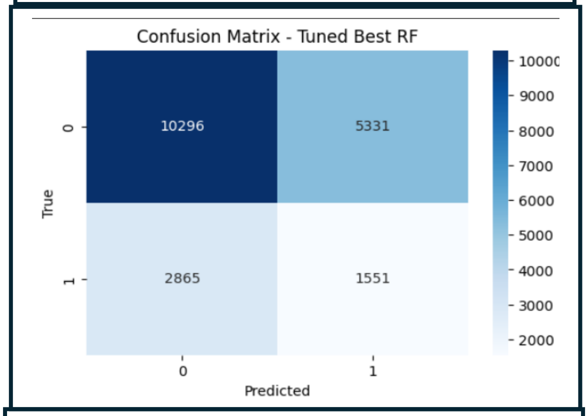
3. k-Nearest Neighbors (k-NN) Classifier
    - After handling class imbalances and hyperparameter tuning
        - Accuracy: 0.6286
        - Precision: 0.2186
        - Recall: 0.2702
        - F1-score: 0.2417



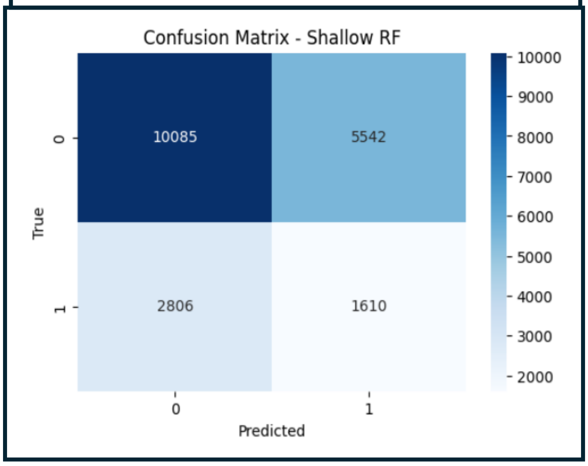
4. Random Forest Classifier
    - Default Random Forest
        - Accuracy: 0.7725
        - F1 Score: 0.0185
        - Precision: 0.1870
        - Recall: 0.0097
        - AUC-ROC: 0.4978



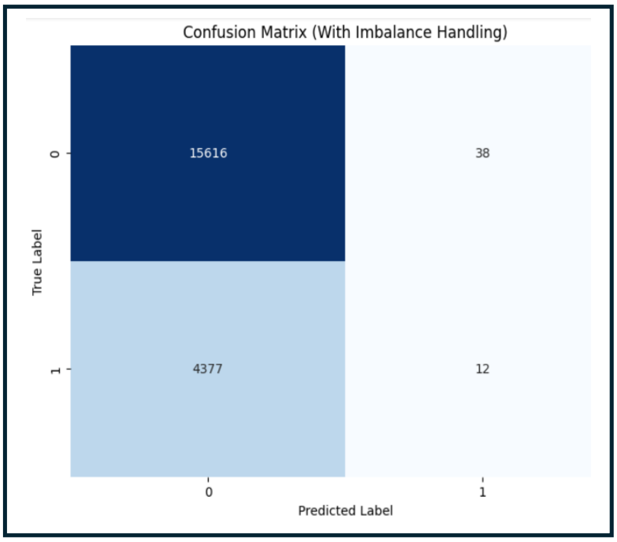
    - Tuned Best Random Forest
        - Accuracy: 0.5911
        - F1 Score: 0.2746
        - Precision: 0.2254
        - Recall: 0.3512
        - AUC-ROC: 0.5050



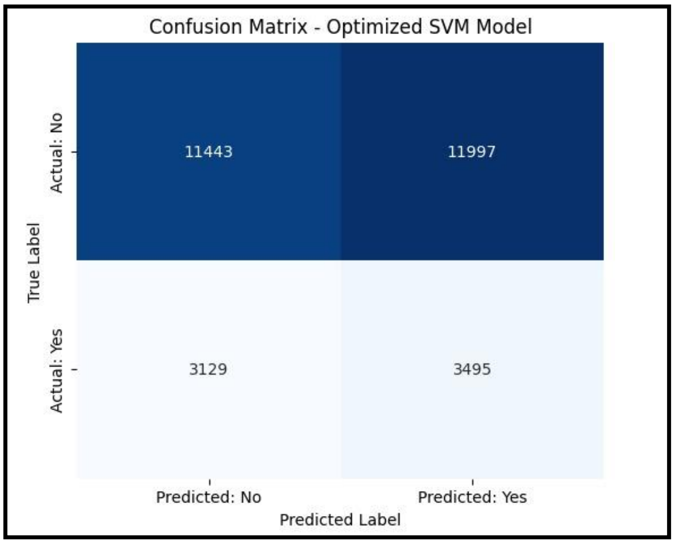
    - Shallow Random Forest
        - Accuracy: 0.5835
        - F1 Score: 0.2784
        - Precision: 0.2251
        - Recall: 0.3646
        - AUC-ROC: 0.5047



5. Naive Bayes Classifier
    - Both with and without handling class imbalances
        - Accuracy: 0.7797
        - Precision: 0.2400
        - Recall: 0.0027
        - F1 Score: 0.0054



6. Support Vector Machine (SVM)
    - After hyper-parameter tuning
    - Accuracy: 0.4969
    - Precision: 0.2256
    - Recall: 0.5276
    - F1 Score: 0.3161



<u>Best Model with highest metrics</u>

- **Tuned SVM**
- It has the highest F1-score (0.3161) and the highest recall (0.5276), meaning it finds the minority class more often than any other model while keeping a reasonable precision (0.2256). For imbalanced problems, F1 and recall are the most informative metrics.
- Metrics: Accuracy 0.4969, Precision 0.2256, Recall 0.5276, F1 0.3161.

6. Ethical Considerations and Bias Mitigation

<u>Ethical Considerations</u>

- **Data Privacy:** Ensure that patient data remains confidential and anonymized; sensitive details such as IDs and dates are excluded from model training.
- **Informed Consent:** Data should be collected and used with consent and in compliance with medical data protection standards.
- **Transparency:** Model assumptions, data preprocessing steps, and limitations are clearly documented to maintain transparency.
- **Accountability:** The results should not be used for real medical decisions without human expert review.
- **Fairness:** Avoid discriminatory outcomes that favor certain patient groups over others.

<u>Bias Mitigation</u>

- **Class Imbalance Handling:** Applied techniques such as class weighting and stratified splitting to reduce bias toward the majority class.
- **Fair Evaluation Metrics:** Used F1 Score and Recall as key metrics since they better reflect performance on the minority (survived) class compared to accuracy.
- **Data Quality Checks:** Removed duplicates, handled missing values, and validated outliers to ensure the dataset represents patients fairly.
- **Model Comparison:** Multiple algorithms (SVM, Naive Bayes, Decision Tree, etc.) were compared to avoid bias from relying on a single model type.
- **Continuous Monitoring:** Future improvements should include periodic retraining and fairness testing as more data becomes available.

7. Reflections and Lessons Learned

- Imbalance is the central challenge; class weighting helped but was not sufficient to maximize performance across all metrics.
- Scale-sensitive models require careful preprocessing and pipeline encapsulation to avoid leakage.
- Hyperparameter tuning substantially affects robustness; randomized searches with stratified folds find reasonable configurations faster than exhaustive grids for large spaces.
- EDA before and after each preprocessing step is essential to validate assumptions.

8. References

- Source : [Lung Cancer Dataset]

- Logistic Regression – Lab Sheet 05 , https://scikit-learn.org/stable/

- K Nearest Neighbors(KNN) - Lab Sheet 05 , Fix, E., & Hodges, J. L. (1951). *"Discriminatory Analysis. Nonparametric Discrimination: Consistency Properties."* USAF School of Aviation Medicine.

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- Naive Bayes Classifier - https://scikit-learn.org/stable/ , Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill.

- Random Forest Classifier - https://scikit-learn.org/stable/ , Breiman, L. (2001). *"Random Forests."* Machine Learning, 45(1), 5–32.

- Concepts on supervised learning algorithms, model evaluation (accuracy, precision, recall, F1), and bias mitigation strategies.
  -- Raschka, S. & Mirjalili, V. (2022). *Python Machine Learning (4th Edition).* Packt Publishing.